

Bellman-Ford

$$O(|E| \cdot |V| + |V|^2)$$

If G is connected, $|E| \geq |V| - 1$. So, the runtime is $O(|E| \cdot |V|)$.

If G is disconnected, modify the algorithm. Find G 's connected components and run the algorithm separately on each.

Denoting the connected components by $(V_1, E_1) \dots (V_r, E_r)$, the running time $\sum_{i \in [r]} O(|V_i|^2) \leq O(|V|) \cdot \sum_{i \in [r]} (|E_i|) = O(|V| \cdot |E|)$

$$\sum_{i \in [r]} O(|V_i| \cdot |E_i|) \leq O(|V|) \cdot \sum_{i \in [r]} |E_i| = O(|V| \cdot |E|)$$

$$\sum_{i \in [r]} (|E_i|) = |E|$$

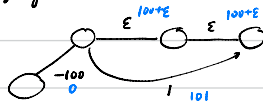
because the components are edge disjoint.

Another way: There is a $c > 1$ st. for every (large enough) $n \in \mathbb{N}$.

$$\sum_{i \in [n]} c |V_i|^2 \leq c |V| \sum_{i \in [n]} |E_i| \leq c |V| |E|$$

Johnson's Algorithm

Why reweighting does not work?



Proof of reweighting lemma:

Let $P = (v_0, \dots, v_k)$ be a path in G

$$w'(P) = \sum_{i \in [k]} w'(v_{i-1}, v_i) = \sum_{i \in [k]} w(v_{i-1}, v_i) + h(v_{i-1}) - h(v_i) \xrightarrow{\text{telescoping}} w(P) + h(v_0) - h(v_k).$$

In particular, for any two paths P, P' , both starting at v_0 , ending at v_k , $w'(P') - w(P') = w'(P) - w(P) \Rightarrow w'(P') - w'(P) = w(P') - w(P)$

$\Rightarrow P$ is shortest under w iff P is shortest under w' . Also, for any cycle C , $w'(C) = w(C)$.

Proof of lemma:

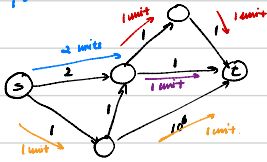
$$w(u, v) + h(u) - h(v) = w(u, v) + \text{dist}(s, u) - \text{dist}(s, v)$$

But we know $\text{dist}(s, v) \leq \text{dist}(s, u) + w(u, v)$ by triangle inequality.

Hence $w(u, v) \geq 0$. $\forall u, v$.

Max-flow

Max: 3 units



Proof of lemma 1:

$$f^{in}(u) = f^{in}(u \setminus \{s\}) = f^{in}(t) + f^{in}(u \setminus \{s, t\}) \quad \forall u \in V \setminus \{s, t\}, f^{in}(u) = f^{out}(u)$$

$$f^{out}(u) = f^{out}(u \setminus \{t\}) = f^{out}(s) + f^{out}(u \setminus \{s, t\}) = f^{out}(s) + f^{in}(u \setminus \{s, t\})$$

$$\Rightarrow f^{in}(u) - f^{out}(u) = f^{in}(t) - f^{out}(s)$$

Now observe:

$$f^{in}(u) = \sum_{v \in V} \sum_{e \in \text{going}} f(e) = \sum_{e \in E} f(e)$$

$$f^{out}(u) = \dots = \sum_{e \in E} f(e)$$

$$\text{Hence, } f^{out}(u) = f^{in}(u).$$